

STANDARD OPERATING PROCEDURE

SOP-BIO-2026-042

Purification and Kinetic Enzymatic Assay of Recombinant Human Kinase V-321

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Department	Molecular Biochemistry & Bioassays	Review Cycle	Bi-annual (Next: Jan 2027)
Target Molecule	Human Kinase V-321 (His ₆ -tagged)	Expression Host	<i>E. coli</i> BL21(DE3) pET-28a(+)
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Quality Assurance	Dr. Sarah Jenkins, QA Compliance Manager	Classification	Controlled Operating Protocol

1. PURPOSE & SCOPE

This Standard Operating Procedure (SOP) outlines the validated workflow for cell lysis, Immobilized Metal Affinity Chromatography (IMAC) purification, size-exclusion buffer exchange, and continuous spectrophotometric NADH-coupled kinase activity assay of human recombinant Kinase V-321.

- Scope:** Applies to all research personnel within Molecular Biochemistry at Meridian Biosciences Inc. executing lot release, stability testing, or kinetic profiling for Project Kinase-V321.
- Quality Goal:** Maintain lot consistency with protein purity $\geq 95\%$, specific activity ≥ 280 U/mg, and assay coefficient of variation (CV) $< 5\%$.

2. SAFETY PRECAUTIONS & HAZARD CONTROL

SAFETY WARNING & HAZARD CONTROL Strict adherence to Biosafety Level 2 (BSL-2) practices is mandatory. All operations involving chemical lysis reagents and inhibitors must be conducted with appropriate Personal Protective Equipment (PPE).

- PMSF (Phenylmethylsulfonyl fluoride):** Highly toxic serine protease inhibitor ($LD_{50} = 200$ mg/kg). Causes severe skin burns. Prepare fresh solutions in anhydrous Isopropanol inside a certified chemical fume hood.
- Acrylamide & SDS (SDS-PAGE):** Neurotoxin and suspected carcinogen. Handle monomer gels in the fume hood.
- Imidazole & Dithiothreitol (DTT):** Corrosive skin irritants. Avoid inhalation of dust or contact with eyes.
- Required PPE:** Lab coat, safety goggles with side shields, double nitrile gloves, and closed-toe non-porous footwear.

3. REAGENTS, MATERIALS & EQUIPMENT

3.1 Chemical Reagents & Solutions

Reagent / Chemical	Grade / Specification	CAS No.	Storage	Safety / Prep Note
Tris-HCl	Mol. Biology Grade ($\geq 99.8\%$)	77-86-1	Room Temp	Buffer base; adjust pH with HCl
Sodium Chloride (NaCl)	ACS Reagent ($\geq 99.0\%$)	7647-14-5	Room Temp	High purity ionic strength agent
Imidazole	ReagentPlus ($\geq 99.0\%$)	288-32-4	Room Temp	Corrosive; light sensitive
Ni-NTA Sepharose High Perf.	5 mL Prepacked Column	N/A	4–8°C	Do not freeze; store in 20% EtOH
Adenosine 5'-triphosphate (ATP)	Disodium Salt ($\geq 98.0\%$)	34369-07-8	-20°C	Prepare 100 mM stock in dH ₂ O
NADH (Reduced NAD ⁺)	Disodium Salt ($\geq 98\%$)	606-68-8	-20°C (Desic.)	Prepare fresh daily; light sensitive
Phosphoenolpyruvate (PEP)	Monopotassium Salt	4265-07-2	-20°C	Prepare 50 mM stock in buffer
PK/LDH Enzyme Suspension	Rabbit Muscle Mix	N/A	2–8°C	Do not freeze or vortex vigorously
Substrate Peptide STK-1	Synthetic ($\geq 98.0\%$ HPLC)	Custom-STK1	-80°C	Resuspend in ultrapure dH ₂ O

3.2 Equipment Specifications

Equipment Type	Model / Manufacturer	Calibration Standard	Operational Criteria
FPLC System	AKTA pure 25 (Cytiva)	Annual IQ/OQ (CAL-014)	Flow rate range: 0.1–25 mL/min
UV-Vis Microplate Reader	SpectraMax M5 (Molecular Dev.)	NIST Traceable Filters	Temp stability: $30.0 \pm 0.2^\circ\text{C}$
Ultrasonic Disruptor	Sonifier SFX-550 (Branson)	Microtip Calibration	Output: 40% amplitude, pulsed
Refrigerated Centrifuge	Avanti J-26S XP (Beckman)	Rotor Cert. (JA-25.50)	Max RCF: $50,000 \times g$ at 4°C

4. REAGENT & BUFFER PREPARATION

Prepare all buffers using Milli-Q ultrapure water (18.2 M Ω ·cm resistivity at 25°C). Filter chromatography buffers through a 0.22 μm PES membrane prior to use.